10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 10 −5 0 5 10

UMAP_1

UMAP_2

0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16

17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25

tSNE_1

tSNE_2

0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16

17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

P2RY12

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

P2RY12

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CX3CR1

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

CX3CR1

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

TREM2

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

TREM2

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

2

4

6

APOE

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

2

4

6

APOE

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

2

4

6

FTL

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

2

4

6

FTL

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

CD83

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

5

CD83

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

NFKB1

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

NFKB1

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CD2

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

CD2

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

SKAP1

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

SKAP1

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

CD14

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

5

CD14

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CD68

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

CD68

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

GFAP

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

5

GFAP

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

AQP4

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

AQP4

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

SLC1A2

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

5

SLC1A2

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

SLC4A4

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

SLC4A4

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

2

4

6

HSP90AA1

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

2

4

6

HSP90AA1

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

SERPINH1

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

5

SERPINH1

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

DNAH9

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

5

DNAH9

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

SPAG17

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

SPAG17

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

VIM

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

5

VIM

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

S100B

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

S100B

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

IL1RAPL1

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

5

IL1RAPL1

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

KIRREL3

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

KIRREL3

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

MRC1

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

5

MRC1

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CD163

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

CD163

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CD163L1

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

CD163L1

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

MSR1

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

MSR1

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

CLDN5

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

CLDN5

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CD38

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

CD38

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

CD79A

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

CD79A

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

2

4

6

IGHM

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

2

4

6

IGHM

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

2

4

6

IGHG1

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

2

4

6

IGHG1

10

4

0

13

6

9 2

20

17

19

3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

−5

0

5

10

− 15 −10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

TSPO

10

4

0

13

6

9

2

20

17

19 3

1

8

5

22

14

26

16

30

24

18

29

7

12

11

21

25

27

31

23

32

28

15

− 25

0

25

− 25 0 25 50

tSNE_1

tSNE_2

0

1

2

3

4

TSPO